

A Connected Cubic-Cover Exclusion at Normal Degrees $(9, 6)$

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Abstract

Let $F, G \in \mathbb{C}[x, y]$ satisfy $\text{Jac}(F, G) = \lambda \in \mathbb{C}^\times$, with $\deg_y F = 9$ and $\deg_y G = 6$. The leading Jacobian equation expresses their leading y -coefficients, after constant rescaling, as h^3 and h^2 for one $h \in \mathbb{C}[x]$. We prove that no such Keller pair exists when h is not a cube in $\mathbb{C}(x)$.

The proof passes to the connected cyclic cover $H^3 = h$, where the μ_3 -character and a triangular approximate-root argument give

$$f = (g^{3/2})_+ + j(g^{1/2})_+.$$

The remaining bracket equations integrate to four weighted first integrals. A centered change of coefficient variables decomposes their level set into two exact branches. One branch has zero Jacobian. On the other, a two-value pole argument excludes every generic level. At the exceptional cuspidal level, a finite-map integrality argument restores polynomial descent and a final valuation identity gives the contradiction.

The degree pair $(9, 6)$ is also the first pair left by a particular repackaging of earlier partial- and total-degree criteria; that arithmetic orientation is prior-theorem bookkeeping, not a new classification result. The theorem does not cover the chart in which h is a cube, and it does not prove the two-dimensional Jacobian conjecture.

1 Introduction and scope

The two-dimensional Jacobian conjecture asks whether a polynomial map

$$(F, G) : \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2$$

with nonzero constant Jacobian is necessarily a polynomial automorphism. The problem remains open. Earlier work gives strong restrictions on the total degrees and Newton data of a possible counterexample; see Guccione–Guccione–Valqui [3], Guccione–Guccione–Horruitiner–Valqui [2, 4], and the references therein.

This paper studies a different, sharply delimited coefficient chart. Write

$$\deg_y F = 9, \quad \deg_y G = 6.$$

The leading Jacobian equation forces the leading coefficients to be coprime powers of one polynomial:

$$[y^9]F = \alpha h^3, \quad [y^6]G = \beta h^2, \quad \alpha, \beta \in \mathbb{C}^\times.$$

The main theorem excludes the case in which h is not a cube in the base rational-function field. Equivalently, it excludes the chart on which adjoining H with $H^3 = h$ gives a connected cubic cover.

The connectedness hypothesis is essential to the argument. It kills the six nontrivial-character constants that otherwise accompany the invariant constant j in the approximate-root expansion. If h is a cube, including when h is a nonzero constant, those six extra constants survive and the lower coefficient system is different. Section 7 records this exact logical boundary.

Relation to earlier results. Moskowitz’s partial-degree criteria [1] and the classical total-degree consequences used there give the arithmetic orientation in Section 2.2. The argument is included to explain why $(9, 6)$, rather than $(8, 6)$, is the first pair not removed by that particular package of criteria. It is a repackaging of prior theorems plus elementary integer enumeration and is not claimed as a new degree theorem. The proof avoids the simultaneous-prime assertion in Lemma 2.3 of [1]: in the present common-leading-factor chart the two reduced progressions coincide, so ordinary Dirichlet suffices.

Approximate roots and intersection constraints for Jacobian pairs have been developed extensively, including in [2]; broader minimal-counterexample and algorithmic frameworks appear in [3, 4]. Karaś’s weighted-bidegree classification [5] provides useful automorphism-side orientation, but it is not used in the connected-chart exclusion. Within a scoped audit of these five sources, no prior theorem was located that states the connected noncube- h , normal-degree $(9, 6)$ exclusion proved here. This is an audit report, not an absolute claim of priority: the literature search was not exhaustive, and independent verification remains necessary before dissemination.

What is not claimed. The theorem neither constructs a counterexample nor proves the plane Jacobian conjecture. It does not eliminate all Keller pairs of normal degrees $(9, 6)$, because the cube- h chart remains. It also does not assert that every hypothetical counterexample can be transformed into the present chart.

2 Statement and arithmetic orientation

2.1 The connected-chart theorem

We use

$$[U, V]_{x,y} = U_x V_y - U_y V_x.$$

Theorem 2.1 (Connected cubic-cover exclusion). *Let $F, G \in \mathbb{C}[x, y]$ satisfy*

$$[F, G]_{x,y} = \lambda \in \mathbb{C}^\times, \quad \deg_y F = 9, \quad \deg_y G = 6.$$

Suppose their leading y -coefficients are

$$[y^9]F = \alpha h^3, \quad [y^6]G = \beta h^2$$

for $h \in \mathbb{C}[x] \setminus \{0\}$ and $\alpha, \beta \in \mathbb{C}^\times$. Then h is a cube in $\mathbb{C}(x)$.

Since $\mathbb{C}$ is algebraically closed, constant rescalings of F, G absorb α, β . We therefore prove the theorem with leading coefficients h^3, h^2 .

2.2 A prior-theorem arithmetic orientation

The following proposition is recorded only to locate the chart. Its automorphism criteria are consequences of the results collected by Moskowicz [1, Theorems 1.1, 1.2, 2.6, and 2.7].

Proposition 2.2 (First survivor of the cited arithmetic sieve). *Let $m = \deg_y F \geq n = \deg_y G > 0$, and apply:*

- (i) *the leading-coefficient relation;*
- (ii) *constant target reductions when $m = n$, and the leading-power target shear when one reduced exponent is one;*
- (iii) *the cited one-coordinate partial-degree criterion; and*
- (iv) *the cited total-degree criteria after a sufficiently high source shear.*

The first degree pair not certified automorphic by this sieve is $(m, n) = (9, 6)$. At that pair the reduced data are

$$(a, b, d, t) = (3, 2, 3, 3).$$

Proof. Let $f_m = [y^m]F$ and $g_n = [y^n]G$. The coefficient of y^{m+n-1} in the Jacobian is

$$nf'_m g_n - mf'_n g'_m = 0.$$

Write

$$d = \gcd(m, n), \quad m = ad, \quad n = bd, \quad \gcd(a, b) = 1.$$

Unique factorization in $\mathbb{C}[x]$, followed by constant rescaling, gives

$$f_m = h^a, \quad g_n = h^b$$

for a nonzero $h \in \mathbb{C}[x]$. Put

$$s = \deg h, \quad t = \gcd(d, s),$$

with $\gcd(d, 0) = d$. The two partial-degree invariants are

$$\gcd(m, \deg f_m) = at, \quad \gcd(n, \deg g_n) = bt.$$

Choose L larger than the x -degree of every coefficient of F, G , and make the source shear $y \mapsto y + x^L$. The resulting total degrees are

$$as + adL = atp, \quad bs + bdL = btp, \quad p = \frac{s}{t} + \frac{d}{t}L.$$

If $s > 0$, the two coefficients in the progression defining p are coprime, so Dirichlet's theorem permits an arbitrarily large L for which p is prime. If $s = 0$, then $t = d$, $p = L$, and one chooses L prime. Consequently the total-degree gcd is tp .

The cited total-degree results certify automorphy when $t = 1$ or $t = 2$. They also certify automorphy if either at or bt lies in

$$\{1, 4\} \cup \{\text{prime numbers}\},$$

because the corresponding total degree is respectively a prime, $4p$, or a product of two primes. These conclusions use one common prime progression only.

For a minimal unresolved pair, a constant linear target change removes $m = n$. If $b = 1$, subtracting a constant multiple of G^a from F lowers the ordered y -degree pair. Hence

$$a > b \geq 2, \quad \gcd(a, b) = 1, \quad t \geq 3, \quad t \mid d.$$

The least possible tuple is

$$(a, b, t, d) = (3, 2, 3, 3),$$

which gives $(m, n) = (9, 6)$. Conversely, its partial invariants are $(9, 6)$, and the high-shear total-degree gcd is $3p$; none belongs to the cited classes. Thus it is the first survivor of this sieve. $\square$

Remark 2.3. Proposition 2.2 does not assert that a Keller pair of degree $(9, 6)$ exists. It is not used to prove Theorem 2.1; it only motivates why that coefficient system is examined.

3 Connected normalization and approximate roots

Assume for contradiction that h is not a cube in $\mathbb{C}(x)$. Put

$$L = \mathbb{C}(x)(H), \quad H^3 = h.$$

Because $\mathbb{C}$ contains the cube roots of unity, this is a connected cyclic cubic extension. Let ζ be a primitive cube root of unity and let its deck generator satisfy

$$\sigma(H) = \zeta H.$$

With $z = Hy$, define

$$f(x, z) = F(x, z/H), \quad g(x, z) = G(x, z/H).$$

Then $f, g \in L[z]$ are monic of degrees nine and six and

$$[f, g]_{x, z} = \frac{\lambda}{H}. \tag{1}$$

The deck action sends z to ζz and fixes f, g .

Translate z over L to kill the degree-five coefficient of g . Its deck character makes the translated variable w satisfy $\sigma(w) = \zeta w$. Write

$$g = w^6 + aw^4 + bw^3 + cw^2 + dw + q. \tag{2}$$

The coefficient characters are:

	a	b	c	d	q
wt	2	3	4	5	6
σ	ζ^2	1	ζ	ζ^2	1.

Lemma 3.1 (Triangular approximate-root form). *After a constant target shear and translation, there is a $j \in \mathbb{C}$ such that*

$$\boxed{f = (g^{3/2})_+ + j(g^{1/2})_+}.$$

Here $(\cdot)_+$ denotes the polynomial part of the distinguished Laurent expansion at $w = \infty$.

Proof. For $0 \leq k \leq 9$, put

$$A_k = (g^{k/6})_+.$$

Each A_k is monic of degree k , so there is a unique triangular expansion

$$f = A_9 + \sum_{k=0}^8 c_k(x) A_k, \quad c_k \in L.$$

The full Laurent series $g^{k/6}$ commutes with g , and its negative part begins in degree -1 . Therefore

$$\deg_w[A_k, g] \leq 4.$$

On the other hand,

$$[c_k A_k, g] = c'_k A_k g_w + c_k [A_k, g],$$

and the first summand has leading term $6c'_k w^{k+5}$. Descending from $k = 8$ to $k = 0$, equation (1) forces every $c'_k = 0$. The constant field of the algebraic function field $L/\mathbb{C}$ is $\mathbb{C}$, so $c_k \in \mathbb{C}$.

The distinguished sixth root of g has leading term w , and hence $\sigma(A_k) = \zeta^k A_k$. Since f is invariant and the coefficients c_k are constants, uniqueness of the triangular expansion forces $c_k = 0$ unless $3 \mid k$. A target shear removes the $c_6 g$ term and a target translation removes c_0 . Writing $c_3 = j$ gives the result. $\square$

Expanding Lemma 3.1 gives

$$\begin{aligned} f = & w^9 + \frac{3}{2}aw^7 + \frac{3}{2}bw^6 + \frac{3(a^2 + 4c)}{8}w^5 + \frac{3(ab + 2d)}{4}w^4 \\ & + \frac{-a^3 + 12ac + 6b^2 + 16j + 24q}{16}w^3 - \frac{3(a^2b - 4ad - 4bc)}{16}w^2 + L_1w + P, \end{aligned} \quad (3)$$

where

$$L_1 = \frac{3a^4 - 24a^2c - 24ab^2 + 64aj + 96aq + 96bd + 48c^2}{128}, \quad (4)$$

$$P = \frac{3a^3b - 6a^2d - 12abc - 2b^3 + 16bj + 24bq + 24cd}{32}. \quad (5)$$

Assigning

$$\text{wt}(a, b, c, d, q; j) = (2, 3, 4, 5, 6; 6)$$

makes every coefficient in (3) weighted homogeneous of the weight complementary to its power of w .

4 The four lower first integrals

Define

$$I_{10} = \frac{3}{128} (3a^5 - 24a^3c - 36a^2b^2 + 32a^2j + 48a^2q + 96abd + 48ac^2 + 48b^2c - 128cj - 192cq - 96d^2), \quad (6)$$

$$I_{11} = \frac{3}{128} (15a^4b - 24a^3d - 72a^2bc - 24ab^3 + 64abj + 96abq + 96acd + 48b^2d + 48bc^2 - 128dj - 192dq), \quad (7)$$

$$I_{12} = -\frac{1}{512} (15a^6 - 132a^4c - 288a^3b^2 + 128a^3j + 192a^3q + 672a^2bd + 336a^2c^2 + 768ab^2c - 512acj - 768acq - 384ad^2 + 72b^4 - 384b^2j - 576b^2q - 1152bcd - 192c^3 + 1536jq + 1152q^2), \quad (8)$$

$$I_{13} = -\frac{1}{128} (15a^5b - 21a^4d - 96a^3bc - 48a^2b^3 + 64a^2bj + 96a^2bq + 120a^2cd + 120ab^2d + 144abc^2 - 64adj - 96adq + 48b^3c - 128bcj - 192bcq - 96bd^2 - 144c^2d). \quad (9)$$

Lemma 4.1 (Exact integration of the lower equations). *Let*

$$R_k = [w^k][f, g]_{x,w}$$

after substituting (2) and (3). Then

$$\begin{aligned} R_4 &= I'_{10}, \\ R_3 &= I'_{11}, \\ R_2 &= I'_{12} + \frac{1}{2}aR_4, \\ R_1 &= I'_{13} + \frac{1}{3}bR_4 + \frac{1}{3}aR_3. \end{aligned}$$

The terminal coefficient is

$$R_0 = dP' - L_1q'.$$

Proof. Differentiate (2) and (3), form $f_xg_w - f_wg_x$, and collect coefficients of w . Substitution of (6)–(9) gives the displayed four identities coefficient by coefficient. The constant coefficient is $dP' - L_1q'$. This is a polynomial identity over $\mathbb{Q}[a, b, c, d, q, j, a', b', c', d', q']$; no division or genericity assumption is used. $\square$

Equation (1) and Lemma 4.1 make all four I_r constant. Their weights give the deck characters

$$\sigma(I_r) = \zeta^r I_r.$$

Connectedness and the constant field $\mathbb{C}$ therefore imply

$$\boxed{I_{10} = 0, \quad I_{11} = 0, \quad I_{12} = K, \quad I_{13} = 0} \quad (10)$$

for one $K \in \mathbb{C}$. The terminal equation is

$$\boxed{dP' - L_1q' = \frac{\lambda}{H}.} \quad (11)$$

5 Exact decomposition of the level set

Make the centered change

$$C = c - \frac{a^2}{4}, \quad D = d - \frac{ab}{2}, \quad Q = q - \frac{b^2}{4}.$$

Direct substitution gives

$$I_{10} = \frac{3}{8}(3C^2a - 12CQ - 8Cj - 6D^2), \quad (12)$$

$$I_{11} = \frac{3}{8}(3C^2b + 6CDa - 12DQ - 8Dj), \quad (13)$$

$$I_{12} = \frac{1}{8}(3C^3 - 3C^2a^2 + 18CDB + 12CQa + 8Caj + 6D^2a - 18Q^2 - 24Qj), \quad (14)$$

$$I_{13} = \frac{1}{16}(18C^2D - 9C^2ab - 6CDa^2 + 24CQb + 16Cbj + 12D^2b + 12DQa + 8Daj). \quad (15)$$

The key factorization is the exact identity

$$\boxed{I_{13} = \frac{9}{8}C^2D - \frac{b}{3}I_{10} - \frac{a}{6}I_{11}.} \quad (16)$$

On the levels (10), it gives $C^2D = 0$.

5.1 The branch $C = 0$

If $C = 0$, equation (12) gives $D = 0$, and

$$I_{12} = -\frac{3}{4}Q(3Q + 4j) = K.$$

Thus Q , being algebraic over the constant field, is constant. Set

$$\rho = w^3 + \frac{a}{2}w + \frac{b}{2}.$$

Substitution in (2) and (3) yields

$$g = \rho^2 + Q, \quad f = \rho^3 + \left(\frac{3}{2}Q + j\right)\rho.$$

Both coordinates are functions of ρ , so their Jacobian is zero, contradicting (1). Hence $C \neq 0$.

5.2 The branch $C \neq 0$

Now (16) gives $D = 0$. Equation (13) then gives $b = 0$, and (12) gives

$$a = \frac{4(3Q + 2j)}{3C}. \quad (17)$$

The remaining invariant is the cubic level

$$\boxed{C^3 = R(Q) := 6Q^2 + 8jQ + \frac{8}{3}K.} \quad (18)$$

Since $b = D = 0$, the original coefficients satisfy

$$d = 0, \quad q = Q, \quad P = 0.$$

Reducing (4) modulo (18) gives

$$L_1 = \frac{S(Q)}{24C}, \quad S(Q) := 21R(Q) + 32(j^2 - K). \quad (19)$$

The terminal equation becomes

$$-L_1 Q' = \frac{\lambda}{H}. \quad (20)$$

6 Exclusion of the two levels

6.1 The generic level $K \neq j^2$

Both b and q are deck-invariant, so

$$Q = q - \frac{b^2}{4} \in \mathbb{C}(x).$$

Cubing (20) and using $C^3 = R(Q)$ gives the base-field identity

$$\boxed{h = -24^3 \lambda^3 \frac{R(Q)}{S(Q)^3 Q'^3}.} \quad (21)$$

The discriminant of the quadratic S is

$$12096(j^2 - K).$$

Thus S has two distinct roots when $K \neq j^2$. Neither is a root of R , since a common root in (19) would give $32(j^2 - K) = 0$.

The terminal equation makes Q nonconstant. Regard it as a rational map $\mathbb{P}_x^1 \rightarrow \mathbb{P}_Q^1$. At least one of the two distinct finite roots α of S has a finite preimage: the single point $x = \infty$ cannot lie above two different target values. At such a preimage x_0 , write

$$\text{ord}_{x_0}(Q - \alpha) = e > 0.$$

Then

$$\text{ord}_{x_0} S(Q) = e, \quad \text{ord}_{x_0} Q' = e - 1, \quad \text{ord}_{x_0} R(Q) = 0.$$

Formula (21) gives h a pole of order $6e - 3 > 0$ at the finite point x_0 , impossible for $h \in \mathbb{C}[x]$. This excludes every generic level.

6.2 The cuspidal level $K = j^2$

Now (18) is

$$C^3 = 6 \left(Q + \frac{2j}{3} \right)^2.$$

On $C \neq 0$, put

$$v = \frac{Q + 2j/3}{C}.$$

Then

$$C = 6v^2, \quad Q = 6v^3 - \frac{2j}{3}, \quad a = 4v, \quad c = 10v^2, \quad b = d = 0, \quad (22)$$

and the normalized pair becomes

$$f = w^9 + 6vw^7 + 21v^2w^5 + 35v^3w^3 + \frac{63}{2}v^4w, \quad (23)$$

$$g = w^6 + 4vw^4 + 10v^2w^2 + 6v^3 - \frac{2j}{3}. \quad (24)$$

A direct calculation gives

$$[f, g]_{v,w} = -567v^6. \quad (25)$$

The depression used in Section 3 occurred over the function field, so polynomial descent of v cannot be assumed. We recover exactly the polynomial quantities needed for the valuation argument.

Lemma 6.1 (Finite-map descent at the cusp). *After substitution back into the original variables,*

$$v^3, vw \in \mathbb{C}[x, y].$$

Proof. Put $g_0 = g + 2j/3$. The pair (f, g_0) is positive weighted-homogeneous for $\text{wt}(v, w) = (2, 1)$. Its only common zero is the origin. Indeed, on the coordinate axes this follows from

$$(f, g_0)|_{v=0} = (w^9, w^6), \quad g_0|_{w=0} = 6v^3.$$

Away from the axes put $T = w^2/v$. A common nonzero zero would make

$$\begin{aligned} T^3 + 4T^2 + 10T + 6 &= 0, \\ T^4 + 6T^3 + 21T^2 + 35T + \frac{63}{2} &= 0, \end{aligned}$$

but the resultant of these two polynomials is $567/8$.

Therefore f, g_0 form a homogeneous system of parameters in the positively graded ring $\mathbb{C}[v, w]$. It follows that $\mathbb{C}[v, w]$ is module-finite over $\mathbb{C}[f, g_0]$, so v^3 and vw are integral over $\mathbb{C}[f, g_0]$.

After returning to x, y , the elements f, g_0 are constant target transforms of the original polynomial coordinates. Hence $\mathbb{C}[f, g_0] \subset \mathbb{C}[x, y]$, and the same monic equations make v^3, vw integral over $\mathbb{C}[x, y]$. Both are deck-invariant and belong to the fixed field

$$L(y)^{\mu_3} = \mathbb{C}(x, y).$$

Since $\mathbb{C}[x, y]$ is normal, integrality now gives $v^3, vw \in \mathbb{C}[x, y]$. □

Write the depressed translation and cusp parameter as

$$w = H(y + s), \quad v = \frac{V}{H}, \quad s, V \in \mathbb{C}(x).$$

The character assertions justify this form: w/H and vH are deck-invariant. By Lemma 6.1,

$$vw = V(y + s) \in \mathbb{C}[x, y],$$

so

$$V \in \mathbb{C}[x], \quad Vs \in \mathbb{C}[x].$$

Also

$$v^3 = \frac{V^3}{h} \in \mathbb{C}[x, y] \cap \mathbb{C}(x) = \mathbb{C}[x],$$

and therefore

$$h \mid V^3. \tag{26}$$

Combining (1), (25), and $v = V/H$ gives

$$\boxed{-189V^6(3hV' - Vh') = \lambda h^3.} \tag{27}$$

Let x_0 be a root of h , and put

$$r = \text{ord}_{x_0} V, \quad s_0 = \text{ord}_{x_0} h.$$

Divisibility (26) gives $s_0 \leq 3r$. If $s_0 \neq 3r$, then the orders of the two sides of (27) are

$$7r + s_0 - 1, \quad 3s_0.$$

Their equality gives $7r - 1 = 2s_0$. Together with $s_0 \leq 3r$, this forces $r = 1, s_0 = 3$, contrary to $s_0 \neq 3r$. Hence $s_0 = 3r$ at every root of h .

A root of V outside the zero set of h would give orders $7r - 1 > 0$ and 0 in (27), also impossible. Thus h and V have the same support, with

$$h = \gamma V^3, \quad \gamma \in \mathbb{C}^\times.$$

But then $3hV' - Vh' = 0$, contradicting (27). This excludes the cuspidal level and completes the proof of Theorem 2.1.

7 The cube- h boundary

Remark 7.1 (Why the proof stops). If h is a cube in $\mathbb{C}(x)$, factorization over $\mathbb{C}$ gives $h = r^3$, up to a constant cube, for some $r \in \mathbb{C}[x]$. This includes all nonzero constant h . Put $z = ry$ and depress g over $\mathbb{C}(x)$. The triangular part of Lemma 3.1 remains valid, but there is no nontrivial deck character. After removing the constant g -multiple and target translation, its exact conclusion is

$$\begin{aligned} f = & (g^{3/2})_+ + \kappa_8(g^{4/3})_+ + \kappa_7(g^{7/6})_+ + \kappa_5(g^{5/6})_+ \\ & + \kappa_4(g^{2/3})_+ + j(g^{1/2})_+ + \kappa_2(g^{1/3})_+ + \kappa_1(g^{1/6})_+, \end{aligned}$$

with seven unrestricted constants at the upper-equation level. In particular, the w^8 -coefficient need not vanish. For instance,

$$F = (xy)^9 + (xy)^8, \quad G = (xy)^6$$

when $h = x^3$, and

$$F = y^9 + y^8, \quad G = y^6$$

when $h = 1$, descend polynomially and retain that coefficient. These examples have zero Jacobian and are not Keller pairs; they only show that descent and the upper equations do not remove the extra constants. The lower cube- h system is outside Theorem 2.1.

No Laurent-time reduction is used in this manuscript. Such a reduction requires additional hypotheses and is best treated separately rather than being folded into the connected theorem.

A Exact-computation companion

The proof is algebraic. A companion script in the research archive,

`current_context/verify_normal_degree_arithmetic_frontier_and_96_connected_reduction.py`,

performs exact symbolic checks of:

- the arithmetic enumeration through $m = 9$;
- the approximate-root expansion (3)–(5);
- the four derivative identities in Lemma 4.1;
- the centered formulas and factorization (12)–(16);
- the generic-level discriminant and the cuspidal parametrization;
- the resultant $567/8$, the cusp Jacobian (25), and the descent identity (27).

The script is a regression check against expansion and sign errors; it does not replace the proof.

Disclosure. AI systems assisted with symbolic-search orchestration, adversarial review, exact-code checking, and draft preparation. The named author is responsible for the mathematical claims, the source, and independent verification before dissemination.

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